

Three Representations Associated with A_5

Class spectra, invariant multiplicities, and direct matrix tests

Computational verification note

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Outcome. All 45 multivariate coefficients indexed by partitions through total degree seven agree in each of the five-dimensional permutation, four-dimensional deleted permutation, and three-dimensional rotation representations. The explicit groups all have order 60. Separate Reynolds projectors confirm the low-degree invariant dimensions.

1 The identity being tested

For a finite matrix group $G \leq GL(V)$, put

$$\mathcal{S}_{G,V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \det(I - t_a g)^{-1}. \quad (1)$$

If g has eigenvalues $\lambda_1, \dots, \lambda_n$, then

$$\det(I - tg)^{-1} = \prod_i (1 - t\lambda_i)^{-1} = \sum_{d \geq 0} h_d(\lambda_1, \dots, \lambda_n) t^d,$$

and $h_d(\lambda(g)) = \chi_{\text{Sym}^d V}(g)$. Hence

$$\begin{aligned} [t_1^{\tau_1} \cdots t_\ell^{\tau_\ell}] \mathcal{S}_{G,V} &= \frac{1}{|G|} \sum_{g \in G} \prod_a \chi_{\text{Sym}^{\tau_a} V}(g) \\ &= \dim \left(\bigotimes_a \text{Sym}^{\tau_a} V \right)^G. \end{aligned} \quad (2)$$

The last equality follows because the group average on the tensor product is the Reynolds projector. Equations (1)–(2) also prove that grouping elements with the same spectrum is exact in every degree.

2 Three class-spectrum formulas

Write $F_{(\lambda_1, \dots, \lambda_n)}(\mathbf{t}) = \prod_{a,i} (1 - \lambda_i t_a)^{-1}$, $\omega = e^{2\pi i/3}$, and $\xi = e^{2\pi i/5}$. The formulas tested are

$$\begin{aligned} \mathcal{S}_{\text{perm}} &= \frac{1}{60} [F_{(1,1,1,1,1)} + 15F_{(1,1,1,-1,-1)} + 20F_{(1,1,1,\omega,\omega^{-1})} \\ &\quad + 24F_{(1,\xi,\xi^2,\xi^3,\xi^4)}], \end{aligned} \quad (3)$$

$$\begin{aligned} \mathcal{S}_{\text{del}} &= \frac{1}{60} [F_{(1,1,1,1)} + 15F_{(1,1,-1,-1)} + 20F_{(1,1,\omega,\omega^{-1})} \\ &\quad + 24F_{(\xi,\xi^2,\xi^3,\xi^4)}], \end{aligned} \quad (4)$$

$$\begin{aligned} \mathcal{S}_{\text{rot}} &= \frac{1}{60} [F_{(1,1,1)} + 15F_{(1,-1,-1)} + 20F_{(1,\omega,\omega^{-1})} \\ &\quad + 12F_{(1,\xi,\xi^{-1})} + 12F_{(1,\xi^2,\xi^{-2})}]. \end{aligned} \quad (5)$$

For (3), a permutation of cycle type (1^5) , $(2^2, 1)$, $(3, 1^2)$, or (5) has precisely the displayed roots of unity as eigenvalues. Removing the always-fixed all-ones line deletes one eigenvalue 1 and proves (4). For (5), a rotation through angle θ has spectrum $(1, e^{i\theta}, e^{-i\theta})$; the rotation counts are 1, 15, 20, 12, 12 for angles $0, \pi, 2\pi/3, 2\pi/5, 4\pi/5$.

The split of the fifth turns is essential in (5): the two classes have different 3D spectra. They may be combined in (3)–(4) because a 5-cycle and its nontrivial powers have the same full permutation spectrum.

3 Independent matrix construction

For the first representation, all 60 even permutations of five symbols are converted directly to 5×5 permutation matrices. For the deleted representation, let B be an orthonormal basis for

$$\{x \in \mathbb{C}^5 : x_1 + \cdots + x_5 = 0\};$$

the matrices are $B^* P_\pi B$. Neither route uses the spectral tables above.

For the rotation representation, start from the twelve icosahedron vertices

$$(0, \pm 1, \pm \varphi), \quad (\pm 1, \pm \varphi, 0), \quad (\pm \varphi, 0, \pm 1), \quad \varphi = (1 + \sqrt{5})/2.$$

Mapping a fixed independent vertex frame to every compatible target frame and retaining the orientation-preserving isometries reconstructs 60 distinct 3×3 matrices.

4 Verification protocol

For every partition τ through total degree seven, the direct route computes h_{τ_a} recursively from the eigenvalues of every explicit matrix and averages $\prod_a h_{\tau_a}$. The formula route uses only the weighted spectra in (3), (4), or (5). It never receives the matrices.

As a third route, the code explicitly constructs each $\text{Sym}^2(V)$ inside $V^{\otimes 2}$ using normalized occupancy states, averages the resulting matrices, and checks projector rank, trace, and idempotence. The same check is also performed on $V \otimes V$.

5 Results

Representation	$ G $	$\dim V$	degree	checks	result
A_5 permutation	60	5	0–7	45	PASS
A_5 deleted permutation	60	4	0–7	45	PASS
A_5 rotation	60	3	0–7	45	PASS

Representation	$[m_{(1)}]$	$[m_{(2)}]$	$[m_{(1,1)}]$	$[m_{(1,1,1)}]$
permutation	1	2	2	5
deleted permutation	0	1	1	1
rotation	0	1	1	1

The symmetric-square projector ranks are respectively 2, 1, 1. Their traces agree with those integers to 3×10^{-16} , and the largest idempotence error is 3.27×10^{-16} . These checks recover the claims that the permutation representation has a fixed line while the deleted and rotation representations do not.

The value $[m_{(1,1,1)}] = 1$ in the rotation case is the invariant orientation tensor in $V^{\otimes 3}$. This is not contradicted by the absence of an invariant cubic in $\text{Sym}^3(V)$: $(1, 1, 1)$ denotes three separate degree-one factors, not the partition (3).

6 Reproducibility and scope

Run the command-line suite from the repository root:

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.a5_representations.verification
```

Open `a5_representations_verification.py` with marimo to change the degree bound and inspect every row.

The finite computation is a transcription and convention test, not the source of the all-degree theorem. All-degree validity follows from (1)–(2) and the complete class spectra; the numerical suite checks the matrix constructions, class sizes, fifth-root choices, and coefficient extraction in representative small dimensions.